

C Programming I

2024 Fall

Midterm

Instructor: Po-Wen Chi

Date: 2024.10.26 PM 2:00-6:00

Policies:

- Online test.
- Do not forget to include your Makefile. TA will only use the command make to build your program. If make fails, you will get zero points and no room for bargaining. **So if you do not know how to solve a problem, please, do not include it in your Makefile.**
- I do not care your source code file names, but the executive binary names should be **mid01, mid02, mid03, mid04, mid05.**
- You can ask TA if you do not understand the problems.

1 Multiplication (20 pts)

Remember that you have developed a program to solve a basic mathematical addition problem in HW01. This time, I want you to develop a multiplication version. The input operand has **one** variable and the product has **two** variables. I promise the variables' names are a, b, c, d . Moreover, **I will not fix the variable location this time.** That is, any digits of the operand can be a variable. For your convenience, I promise all variables will be in $[0 - 9]$. Please list all possible solutions.

```
1 $ ./mid01
2 Please enter the first operand: a11
3 Please enter the second operand: b22
4 Please enter the product          : cd042
5 Solutions:
6 1. a = 0, b = 8, c = 0, d = 9
7 2. a = 1, b = 6, c = 6, d = 9
8 3. a = 2, b = 4, c = 8, d = 9
9 4. 311 x 222 = 69042
10 5. 411 x 022 = 9042
11 6. 911 x 022 = 20042
```



Figure 1: Colorful Card Sequence. The color sequence will be red, yellow, green, blue and then repeated.

If there is any invalid input, print an error message and terminate your program. If there is no possible solutions, print "No solutions" and terminate your program.

2 Colorful Card Sequence (20 pts)

In this problem, I want you to display a colorful card sequence as shown in Fig. 1.

You will make the user input the card size and the number of cards. An example is shown in Fig. 2. For your simplicity, the width should be in $[7 - 40]$, the height should be in $[4 - 10]$ and the number should be in $[1, 40]$. For any invalid inputs, print an error message and terminate the program.

3 Parallelogram (20 pts)

You know what **Parallelogram (平行四邊形)** is, right? This time, I want you to develop a library for Parallelogram related operations.

```

1 // Setup three points for a parallelogram.
2 // Success : return 1; Fail: return 0
3 int32_t setP1( int32_t x, int32_t y);
4 int32_t setP2( int32_t x, int32_t y);
5 int32_t setP3( int32_t x, int32_t y);
6
7 // Check if the current setting is a valid parallelogram .
8 // Yes: return 1; No: return 0
9 int check( void );
10
11 // Return the perimeter of the parallelogram.

```

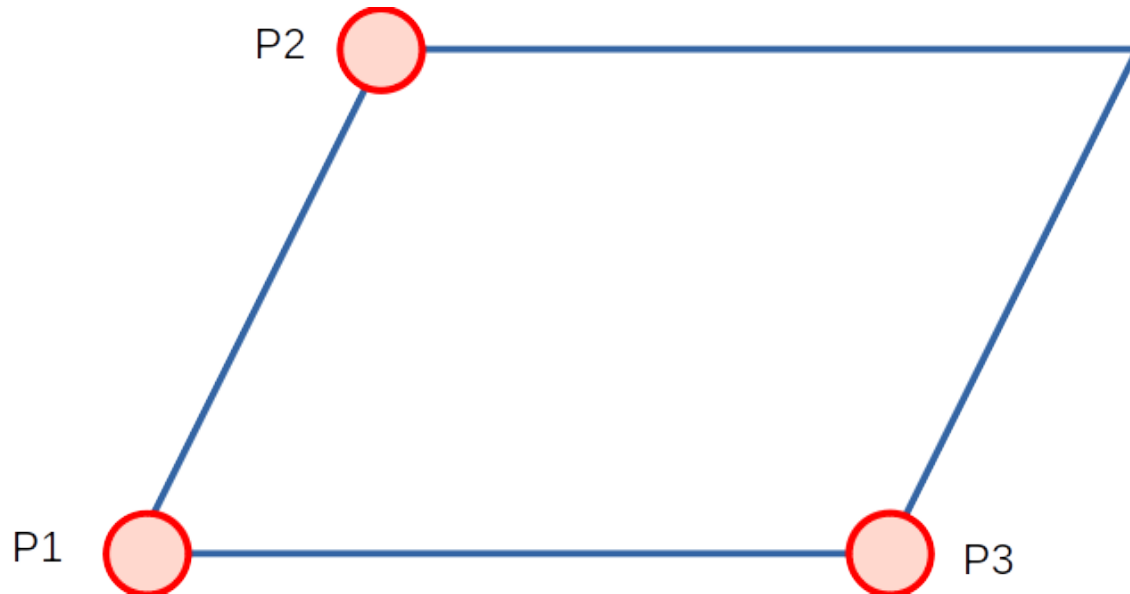



Figure 3: Parallelogram input example.

4. **Turn Right:** The user should enter the degree (0-360) and change the car direction clockwise according to the current direction.
5. **Turn Left:** The user should enter the degree (0-360) and change the car direction counterclockwise according to the current direction.

Once the car reaches the destination, print "**Arrived**" and terminate the program. Since there is the precision problem in C, we define that the car reach the destination if the distance between the car and the destination is less than **0.5 km**. I promise that our TAs will not intentionally use a test case will precision boundary. For any invalid inputs, print a warning message and return to the control panel. The program interface is at follows.

```

1 $ ./mid04
2 Please enter the destination (x,y): (10,10)
3 =====
4 Car is at (0,0)
5 Control Panel:
6 1) Parking
7 2) Drive
8 3) Reverse
9 4) Turn Right
10 5) Turn Left
11 Your Choice: 1
12 =====
13 Car is at (0,0)
14 Control Panel:
15 1) Parking
16 2) Drive
17 3) Reverse
18 4) Turn Right
19 5) Turn Left

```

```

20 Your Choice: 2
21 Velocity (km/hr): 5
22 Duration (hr): 2
23 =====
24 Car is at (0,10)
25 Control Panel:
26 1) Parking
27 2) Drive
28 3) Reverse
29 4) Turn Right
30 5) Turn Left
31 Your Choice: 3
32 Velocity (km/hr): 5
33 Duration (hr): 2
34 =====
35 Car is at (0,0)
36 Control Panel:
37 1) Parking
38 2) Drive
39 3) Reverse
40 4) Turn Right
41 5) Turn Left
42 Your Choice: 4
43 Degree: 45
44 =====
45 Car is at (0,0)
46 Control Panel:
47 1) Parking
48 2) Drive
49 3) Reverse
50 4) Turn Right
51 5) Turn Left
52 Your Choice: 2
53 Velocity (km/hr): 14.14214
54 Duration (hr): 1
55 Arrived

```

You should use **double** for location, velocity, duration in this problem. Choice and degree should be **int32_t**.

5 Regular Expression(20 pts)

Panda is currently studying Automata, but he is really bad at regular expression, he want to know if he have a regular expression, what sequence looks like will accept with the given regular expression. Your task is given the regular $((01^*(2 \cup 3)^*4) \cup (10^*4^*2))^*$, and you will input a sequence, that might be very long, and input -1 mean end of sequence, you should output "accept" or "reject".

```

1 $./mid05
2 Please enter now symbol: 0
3 Please enter now symbol: 1
4 Please enter now symbol: 1

```

```

5 Please enter now symbol: 2
6 Please enter now symbol: 3
7 Please enter now symbol: 4
8 Please enter now symbol: -1
9 this sequence is accepted.

```

explanation:

consider with $(01^*(2 \cup 3)^*4)$

$0 \rightarrow 0$

$11 \rightarrow 1^*$

$2 \rightarrow 2 \cup 3$

$3 \rightarrow 2 \cup 3$

$4 \rightarrow 4$

```

1 $./mid04
2 Please enter now symbol: -1
3 this sequence is accepted.

```

explanation:

because $((01^*(2 \cup 3)^*4) \cup (10^*4^*2))^*$ with a star

that mean any number of $((01^*(2 \cup 3)^*4) \cup (10^*4^*2))$ will be accepted
therefore the empty set will be accepted

```

1 $./mid04
2 Please enter now symbol: 0
3 Please enter now symbol: 1
4 Please enter now symbol: 4
5 Please enter now symbol: 1
6 Please enter now symbol: 0
7 Please enter now symbol: 4
8 Please enter now symbol: -1
9 this sequence is rejected.

```

explanation:

this is not matching neither $(01^*(2 \cup 3)^*4)$ nor $((10^*4^*2))$, so it will be rejected.

you said you don't know what is regular expression? don't worry, I can teach you.

5.1 Regular Expression

A regular expression is a tool used to describe text patterns. It helps us quickly find or match specific strings or symbols, such as finding all words that start with "1" or strings that contain numbers. don't worry, you don't need to process character, there have only number there have some symbol in regular expression, see Table 1

Hint:

1. It is very similar to your homework 2.2.
2. you can make a subset of them.
3. if only done $(01^*(2 \cup 3)^*4)$ or $((10^*4^*2))$, I will give part of points
4. I promise that I will only input a 32-bit integer as a symbol.
5. there have only $[0 \ 4]$ in my regular expression

Symbol	Meaning	Example	Explanation
*	Any number of the symbol	1^*	Any number of "1" is accepted, e.g., 1, 11, or empty string
$\cup$	Union of two regular	$12 \cup 3^*$	Either "12" or "3*" is accepted
()	a sub regular	$1(12)^*$	The string starts with "1" followed by zero or more occurrences of "12", e.g., "1", "112", "11212", etc.

Table 1: Regular Expression Symbols

6 Bonus: Your Comments (5 pts)

Your comments about this class. Any comments are welcomed. Do not worry about typos or any grammar errors. However, you will get nothing if you leave this question blank.